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Climacteric fruits: In most of the plants, there is sharp rise in respiration rate near the end of the development of fruit, called climacteric rise. Such fruits are called climacteric fruits. The ripening on demand can be induced in these fruits by exposing them to normal air containing about 1 ppm of ethylene. A liquid called ethephon is being used in fruit ripening as it continuously releases ethylene.

Example: Tomato, Apples, Banana, Mango.

Non climacteric fruits: All fruits cannot be ripened by exposure to ethylene. Such fruits are called non-climacteric fruits and are insensitive to ethylene.

Example: Grapes, Watermelon, Orange.

15.2.5 Abscisic Acid (ABA) (Stress Phyto Hormone)

1. Discovery

In 1963, the hormone was first isolated by **Addicott *et al.,*** from young cotton bolls and named as **Abscission II**. Eagles and Wareing during 1963–64 isolated a dormancy inducing substance from leaves of *Betula* and called it as dormin. In 1965, it was found by Cornsforth *et al.*, that both dormin and abscission are chemically same compounds and called **Abscisic Acid (ABA)**.

2. Occurrence

This hormone is found abundantly inside the chloroplast of green cells.

3. Precursors

The hormone is formed from mevalonic acid pathway or xanthophylls.

4. Transport in plants

Abscisic acid is transported to all parts of the plant through diffusion as well as through phloem and xylem.

5. Chemical structure

It has carotenoid structure.

6. Bioassay (Rice Coleoptile)

The inhibition of IAA induces straight growth of rice seedling coleoptiles.

7. Physiological effects

- It helps in reducing transpiration rate by closing stomata. It inhibits K^+ uptake by guard cells and promotes the leakage of malic acid. It results in closure of stomata.
- It spoils chlorophylls, proteins and nucleic acids of leaves making them yellow.
- Inhibition of cell division and cell elongation.
- ABA is a powerful growth inhibitor. It causes 50% inhibition of growth in Oat coleoptile.
- It induces bud and seed dormancy.
- It promotes the abscission of leaves, flowers and fruits by forming abscission layers.
- ABA plays an important role in plants during water stress and during drought conditions. It results in loss of turgor and closure of stomata.
- It has anti-auxin and anti-gibberellin property.
- Abscisic acid promotes senescence in